

Scope of Work for the project ‘Short-Term Load Forecasting (STLF) for SLDC Uttarakhand for operational purpose for three (3) years’.

Introduction

Economically efficient generation scheduling and secure grid operation require accurate demand and power forecasting, making Short-Term Load Forecasting (STLF) a critical function for power system operation, load management, and planning. STLF enables utilities and SLDC to estimate future demand and better power forecasting accuracy of renewable energy sources including distributed energy resources with high accuracy, thereby supporting economic generation scheduling, unit commitment, optimal dispatch of generating units, system reliability, and efficient utilization of available resources. Forecasting of total load demand is essential for operational planning, power procurement, reserve management, and maintaining grid stability.

SLDC is responsible for developing methodologies and mechanisms for Daily, Weekly, Monthly, and Yearly demand estimation in terms of MW, MVar, and MWh. Based on the projected demand and the estimated availability of power from various generation sources, SLDC plans and coordinates demand management measures such as load management, power regulation, and controlled power cuts, wherever required, to maintain overall grid security and reliability. Distribution licensees are required to comply with these demand management directives and maintain historical demand databases to support accurate forecasting and operational planning.

Short-Term Load Forecasting is influenced by multiple dynamic factors including historical load patterns, geographical topology, weather conditions, seasonal variations, target day classification (weekday, weekend, holiday, festival, etc.), planned or forced outages, renewable generation variability, and availability of generating resources. Advanced forecasting systems leveraging AI/ML techniques can effectively analyze these parameters to improve forecasting accuracy, support real-time grid operations, minimize operational inefficiencies, and facilitate data-driven decision-making for modern power system management.

The objective of this scope is to engage an Implementation Agency for the design, development, deployment, operation, and maintenance of an integrated, online, web-based Demand and Power Forecasting Solution for SLDC. The proposed solution shall enable accurate and reliable demand and Power forecasting to support efficient grid operations, scheduling, renewable energy integration, and real-time decision-making at the State

Load Dispatch Centre (SLDC). The selected bidder shall provide a comprehensive end-to-end platform comprising advanced forecasting engines, dashboards, analytics modules, visualization tools, summaries, alerts, and reporting capabilities, integrated with relevant data sources and operational systems. The solution is expected to support multiple forecasting horizons, improve forecasting accuracy, enhance grid reliability, reduce operational inefficiencies and DSM penalties, and facilitate data-driven operational planning through a secure, scalable, and user-friendly web-based application architecture.

Key requirements of the proposed solution include forecasting and analytics at 15-minute block-wise granularity to support accurate demand and power estimation, scheduling, and grid operation functions at the SLDC. The system should be scalable and future-ready to align with evolving regulatory requirements, including the anticipated transition from a 15-minute to a 5-minute scheduling and dispatch framework. The solution must support real-time data acquisition, processing, validation, and analytics for operational decision-making and improved forecasting accuracy. Additionally, the platform should ensure high standards of cybersecurity through secure architecture, encrypted communication, audit trails, and robust role-based access control mechanisms to enable controlled and authorized access for different categories of users and stakeholders.

Load Forecasting Module Feature

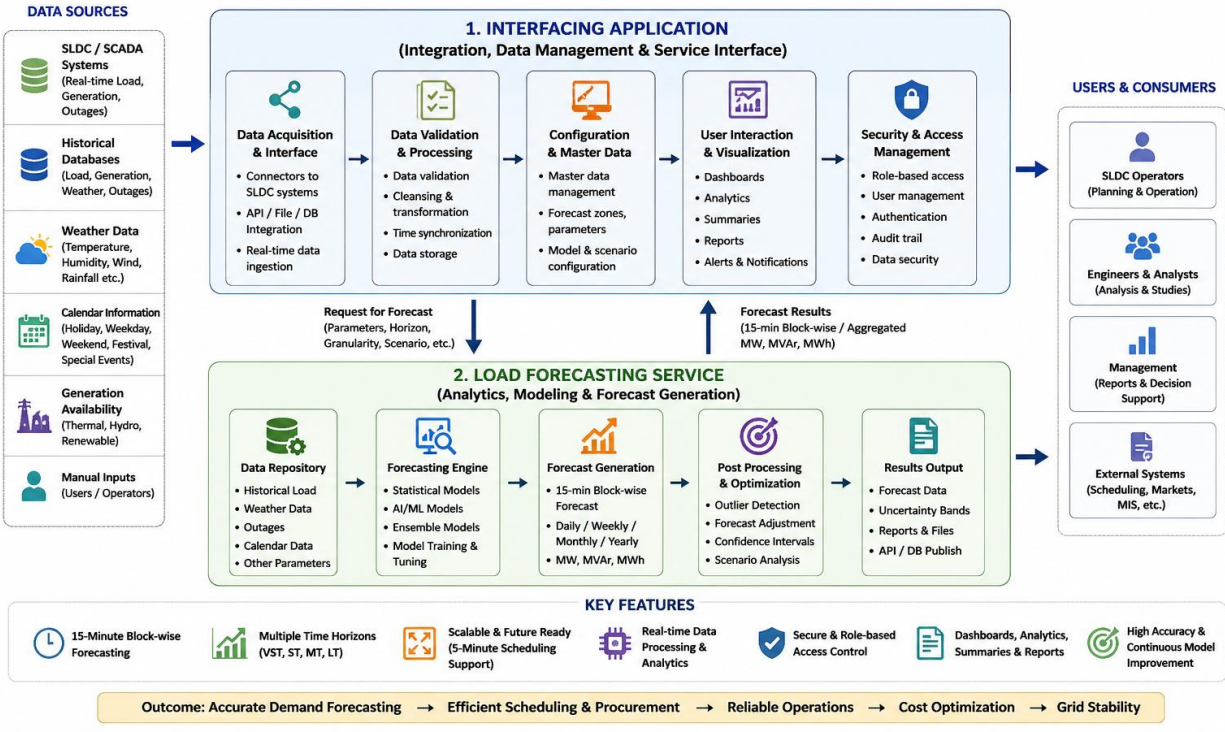
The functionalities of the proposed Load Forecasting Solution are broadly divided into two major applications: The Interfacing Application and the Load Forecasting Service. The Interfacing Application acts as the integration and communication layer between users, SLDC operational systems, and the forecasting engine. It is responsible for collecting, validating, and processing input data from various sources such as SCADA systems, historical databases, weather data providers, scheduling systems, and user inputs. The application facilitates seamless data exchange, user interaction, configuration management, visualization, dashboards, reports, and operational workflows required by the State Load Dispatch Centre Uttarakhand.

The Load Forecasting Service functions as the core analytics and forecasting engine of the solution. It processes the collected data using statistical, analytical, and AI/ML-based forecasting models to generate accurate demand forecasts across multiple time horizons such as short-term (up to one year on rolling basis) forecasting. The service performs demand prediction, trend analysis, anomaly detection, scenario analysis, and forecasting model optimization while supporting real-time and block-wise forecasting requirements. Together, both applications ensure reliable forecasting, efficient operational planning, enhanced decision support, and secure integration with SLDC systems for effective grid management and scheduling operations.

Load Forecasting Solution – Functional Architecture

Functionalities of load forecasting tools are divided among two applications.

Interfacing Application gets required information from users & SLDC applications and provides the service.



Interface Web Application

The Interfacing Web Application shall serve as the primary user interaction, integration, and operational management layer of the proposed Load Forecasting Solution for State Load Dispatch Centre Uttarakhand. The application shall provide a secure, user-friendly, configurable, and web-based platform to support forecasting operations, analysis, reporting, data integration, and operational decision-making. The key functional requirements are detailed below:

a) User-Friendly Web-Based Interface

The application shall provide an intuitive and user-friendly web interface for:

- Forecast generation and monitoring
- Data analysis and visualization
- Dashboard management
- Reports generation
- Forecast configuration and administration
- Scenario analysis and operational planning

The interface should support role-based access for operators, analysts, administrators, and management users.

b) Multi-Horizon Forecasting Support

The application shall support target date and horizon selection as defined under applicable IEGC/SGC/other relevant regulations of UERC/CERC, as the case may, including for:

- Intra-day forecasting
- Day-ahead forecasting
- Weekly forecasting
- Monthly forecasting
- Yearly forecasting on rolling basis.

The solution shall enable flexible forecasting intervals and configurable forecasting horizons based on operational requirements.

c) Configuration of Dependent Parameters

The application shall provide configurable inputs for various demand-influencing parameters, including:

- Planned and forced power outages
- Generation availability
- Hydro and renewable generation status
- Holidays and special events
- Weather parameters such as:
 - Temperature
 - Humidity
 - Rainfall
 - Wind conditions
- Seasonal load variations
- Industrial and commercial load patterns
- Festival-related demand changes

The system shall support operator-defined configuration and modification of these parameters.

d) Real-Time Forecast Revision Capability

The application shall support dynamic revision of demand anticipation based on:

- Real-time power system conditions
- Updated weather information

- Grid disturbances
- Actual load trends
- Generation availability changes

Forecasts should be revisable after every time block to improve forecasting accuracy and operational responsiveness.

e) Historical SCADA Data Upload

The application shall provide mechanisms to:

- Upload historical SCADA data
- Validate and preprocess imported data
- Store and archive historical datasets
- Transfer the processed data to the Load Forecasting Service

The solution shall support bulk data upload and automated data synchronization.

f) Real-Time Data Acquisition

The application shall support acquisition and transmission of real-time operational data for current-day forecasting, including:

- Real-time load data
- Generation data
- Frequency and system parameters
- Outage status
- Renewable generation information

The application shall interface with SCADA/EMS systems and provide data feeds to the forecasting engine as required.

g) Forecast Visualization & Accuracy Analytics

The application shall provide visualization tools for comparison and analysis of:

- Forecasted demand
- Actual demand
- Historical trends
- Forecast deviation patterns

The system shall support graphical trend analysis and display multiple forecasting accuracy parameters, including:

- Bias variation
- Mean Absolute Percentage Error (MAPE)

- Mean Square Error (MSE)
- Percentage deviation
- Percentage accuracy
- Forecast confidence indicators

h) Manual Editing of SCADA Data

During periods when SCADA data is unavailable, frozen, or corrupted, the application shall provide authorized users with the ability to:

- Manually edit SCADA data
- Correct erroneous values
- Validate modified data
- Approve data before submission to the STLF module

All modifications shall be audit logged.

i) Forecast Data Archival

The application shall maintain a centralized archival repository for:

- Forecasted demand data
- Historical forecast runs
- Actual demand records
- Forecast revisions
- Accuracy analysis reports

The archival mechanism shall support long-term storage and retrieval.

j) Configurable Reporting Framework

The application shall provide configurable reporting capabilities for:

- Daily reports
- Weekly reports
- Monthly reports
- Seasonal and annual reports
- Exception and deviation reports

Operators shall be able to:

- Configure report formats
- Define report templates
- Select parameters and KPIs
- Export reports in multiple formats

The reporting framework should not be hardcoded and must support future customization.

k) Integration with Scheduling Systems

The application shall support integration with:

- SLDC scheduling modules
- NRLDC scheduling portal
- External scheduling and reporting systems

The system shall automatically fetch:

- Schedules
- Generation allocation data
- Drawal schedules
- Interchange schedules

Standard integration mechanisms such as APIs, web services, and secure data exchange protocols shall be provided.

l) SCADA Report Module Integration

The application shall integrate with the SCADA reporting system to automatically retrieve:

- Historical operational data
- Real-time system data
- Load curves
- Generation data
- Grid parameters

The application shall securely exchange this data with the Load Forecasting Service whenever required.

m) Weather Data Integration

The application shall support:

- Acquisition of weather forecast data from IMD/other reputed forecasting agencies
- Recording and archival of weather parameters
- Correlation analysis between weather and demand

Weather parameters shall include:

- Temperature
- Humidity
- Rainfall
- Wind conditions

- Seasonal weather patterns

The system shall analyze the impact of weather on load behavior for improved forecasting accuracy.

Forecast Accuracy Monitoring

Forecast accuracy shall be continuously monitored and evaluated.

The system shall:

- Highlight forecast deviations exceeding predefined thresholds
- Provide color-coded visualization of forecast accuracy
- Allow user-configurable deviation bands
- Generate alerts for significant variations

The anticipated demand variation should ideally remain within $\pm 1\%$ of actual demand values.

a) Merit Order Dispatch (MOD) Interface

The application shall provide operators with the capability to:

- Define Merit Order Dispatch (MOD)
- Configure generation priorities
- Set scheduling preferences
- Define operational constraints

Based on these inputs, the system shall recommend operational measures to minimize deviation between:

- State demand
- State availability
- Scheduled generation

b) Scheduling & Optimization Recommendations

The application shall support operational optimization by:

- Accommodating MOD principles
- Recommending schedule revisions
- Suggesting surrender of allocations
- Advising short-term market purchases
- Optimizing procurement strategies

These recommendations shall support:

- Short-term planning up to one year on rolling basis

The solution shall assist SLDC operators in achieving economical and reliable grid operation while minimizing operational risks.

Features of Load Forecasting Module as a Service

The Load Forecasting Module as a Service shall function as the core forecasting and analytics engine of the proposed Demand Forecasting Solution. The module shall utilize advanced statistical techniques, machine learning, artificial intelligence algorithms, historical operational patterns, weather analytics, and real-time system data to generate accurate forecasts for demand and state power availability across multiple forecasting horizons. The key functional requirements of the module are as follows:

1. Short-Term Load Forecasting (STLF)

The STLF module shall be responsible for forecasting on different time horizons including but not limited to the time horizons defined under applicable IEGC/SGC regulations/other UERC or CERC regulations and amendments of these regulations:

- State power demand
- State power availability at 5-minute / 15-minute resolution for:
 - Current day (D)
 - Next day (D+1)
 - Weekly forecasting
 - Monthly forecasting
 - Yearly forecasting on rolling basis.

The module shall support:

- Block-wise forecasting
- Real-time forecast updates
- Dynamic revision of forecasts
- Intra-day forecasting adjustments
- Seasonal demand analysis
- Trend analysis
- Resource planning
- Procurement planning
- Capacity planning support

The forecasting engine shall assist the State Load Dispatch Centre Uttarakhand in scheduling, dispatch planning, grid balancing.

2. Automated Forecast Execution

The forecasting modules shall:

- Receive forecasting requests from the Interfacing Application

- Identify the required forecasting horizon
- Execute the corresponding forecasting model automatically
- Return forecast results to the Interfacing Application

The system shall support seamless interaction between:

- STLF modules
- User interfaces and dashboards

3. Data Integration & Input Handling

The forecasting service shall support receipt and processing of:

- Historical operational data
- Real-time SCADA data
- Scheduling data
- Generation schedules
- Weather information
- Configuration parameters
- Outage information
- Market and operational inputs

The module shall dynamically receive and process data from the Interfacing Application as and when required for forecasting operations.

4. Secure Cloud-Based Deployment

The selected bidder shall provide, provision, host, manage, operate, secure, and maintain the complete application, database, analytics, AI/ML platform, storage, backup, disaster recovery, and associated infrastructure on a MeitY-empanelled cloud service provider. The cloud infrastructure shall be arranged entirely by the bidder, and all associated subscription, licensing, hosting, storage, security, backup, disaster recovery, and operational costs shall be included in the bidder's scope and commercial proposal.

The Load Forecasting Module shall be deployed on a secure cloud infrastructure with appropriate cybersecurity controls and operational safeguards.

The solution shall include:

- Secure hosting environment
- Data encryption
- Role-based access control
- Audit logging

- Secure APIs and communication protocols
- Backup and disaster recovery mechanisms
- High availability architecture

State Load Dispatch Centre Uttarakhand shall have the right to request and review all relevant security compliance documents, certifications, and deployment architecture details to validate the security and reliability of the cloud environment.

5. Weather Data Acquisition & Integration

The forecasting module shall acquire weather forecast data from:

- India Meteorological Department (IMD)
- Equivalent authorized weather institutions

The system shall incorporate weather parameters such as:

- Temperature
- Humidity
- Rainfall
- Wind conditions
- Seasonal weather trends

Weather analytics shall be integrated into forecasting models to improve forecast accuracy and enhance demand prediction capabilities.

6. Weather-Based Demand Correlation Logic

The forecasting engine shall include analytical logic to study and model the impact of weather variations on electricity demand behavior.

The system shall:

- Analyze historical demand-weather relationships
- Identify weather-sensitive load patterns
- Model seasonal and regional demand variations
- Continuously improve forecasting models based on observed behavior

Given that power demand is highly dynamic in nature, the forecasting methodology shall be adaptive and continuously optimized based on:

- Forecast accuracy performance
- Changing consumption behavior
- Operational trends
- Weather variability

- Renewable integration patterns

The forecasting algorithms and methodologies shall support periodic revision, retraining, and enhancement to ensure sustained forecasting accuracy and operational effectiveness over time.

Generation Forecasting Module Features

The generation Forecasting Module shall provide integrated generation Forecasting and Weather Forecasting services for renewable and conventional energy systems to support grid operations, scheduling/re-scheduling and regulatory compliance.

The module shall include:

- Yearly on rolling basis, monthly, weekly, Day-ahead, intraday, and real-time forecasting for Solar, DERs, Hydro etc.
- 15-minute / 5-minute block-wise generation forecasting
- AI/ML and weather-based forecasting models
- Forecast revision and schedule management
- Forecast vs actual analysis
- Automated schedule preparation and SLDC/RLDC schedule integration
- Historical data analysis and forecasting accuracy tracking
- Curtailment and performance analytics
- Solar irradiance (GHI, DNI, DHI)
- Temperature, humidity, pressure, rainfall, and cloud cover
- Nowcasting, day-ahead, and weekly weather forecasting
- Severe weather alerts and weather analytics

Proposed Manpower

The Service Provider shall deploy at least one (1) qualified full-time Resident Engineers at the SLDC Office, Dehradun in rotating shift to provide 24×7×365 onsite support services for the operation, maintenance, enhancement, and smooth functioning of the Integrated Software Solution. The scope of support shall include, but not be limited to, Load Forecasting, Renewable Energy (RE) Forecasting, mapping and integration of Distributed Energy Resources (DERs) with the Forecasting System, system administration, user support, data management, report generation, and all other associated day-to-day operational activities.

The Service Provider shall ensure uninterrupted 24×7×365 onsite support throughout the contract period by deploying adequate and competent manpower **across all shifts**, including weekends, Gazetted Holidays, and other non-working days. The staffing and deployment plan shall be structured such that **one Resident Engineer** is physically available at SLDC Office, Dehradun **during each shift, thereby ensuring continuous round-the-clock operational support through the deployment of**

three Resident Engineers. The Service Provider shall further maintain adequate backup resources and an escalation mechanism to promptly address critical incidents, system failures, enhancement requirements, peak operational workloads, and any other unforeseen exigencies.

The deployed Resident Engineers shall be responsible for continuous operational support, system administration, proactive monitoring of applications and infrastructure, performance monitoring, troubleshooting and resolution of technical issues, user assistance, stakeholder coordination, data validation, configuration management, knowledge transfer, and support for system upgrades, patches, and enhancements. The Resident Engineers shall also assist SLDC in operational planning, forecasting activities, system utilization, and effective use of the solution.

The Service Provider shall ensure that the deployed manpower possesses the requisite qualifications, experience, and domain expertise necessary for carrying out the assigned responsibilities. Any replacement of Resident Engineers shall be subject to prior approval of SLDC and shall be made with personnel having equivalent or higher qualifications and experience.

The Service Provider shall be responsible for maintaining the prescribed Service Levels and ensuring high availability, reliability, security, performance, and operational continuity of the solution throughout the tenure of the contract.

Proposed Manpower Experience

Minimum Qualification -B.E./ B. Tech./BCA / MCA / or equivalent or higher degree.

Min. Experience: Experience of more than Three (3) years of experience of using AI/ML tools for Load Forecasting & RE Forecasting in Govt./CPSU/State transmission Utility/State Distribution Utility/Private Discom/utilities etc.

Payment Terms:

1. 40% of the development charges (S. No. 1 of the BOQ in Annexure-1) shall be paid after successful completion of trial run of the final version of the STLF module (incorporating all necessary change requests and customization) against the completion certificate of Site Acceptance Test (SAT) issued by the Engineer In-Charge of the project.
2. Balance 30% of the development charges (S. No. 1 of the BOQ in Annexure-1) shall be paid after declaration of Go-Live date by the Engineer In-Charge.
3. 30% of the payment for item mentioned at S. No. of the BOQ (Annexure-1) shall be paid after completion of 1-year Warranty period from the Go-Live date of the STLF Module.
4. Payment of AMC services (mentioned at S. No. 2 & 3 of Annexure-1) shall be done on pro-rata Quarterly basis after end of the quarter.
5. Payment of Cyber Security Audit (mentioned at S. No. 4, 5 & 6 of Annexure-1) shall done after providing relevant certificates of Cyber Security Audit carried out.
6. Payment of Cloud Server hosting (mentioned at S. No. 7, 8 & 9 of Annexure-1) shall be done on yearly basis at the start of each year.
7. Payment of AMC services (mentioned at S. No. 10, 11 & 12 of Annexure-1) shall be done on pro-rata Quarterly basis after end of the quarter.